Data-Driven Approach to Flood Control Monitoring and Forecasting

A Capstone Project
Presented to the Faculty of the
College of Computer Studies
Naga College Foundation, Inc.
Naga City

In Partial Fulfillment of
the Requirement for the Degree
Bachelor of Science in Information Systems

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2027

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1 Introduction

1.1 Background of the Study

Many natural disasters and occurrences significantly affected communities due to climate change. One of these is flooding, which has become prevalent in recent years and is considered a major global problem. The number of people exposed to floods globally has steadily risen from 28.1 million in 1970 to 35.1 million in 2020—an increase of 24.9%, with most flood-related deaths and economic losses recorded in Asia [1]. This increase in frequency of occurrence causes disruption of services, damages, and in some cases even costs lives. Climate change and urban development have intensified flooding threats, as water flow becomes constrained by man-made structures [2].

The Philippines' geographical position makes it particularly susceptible to natural disasters, and flooding has become more frequent and devastating in recent years. Tropical Storm Trami (Kristine) exemplifies this severity, bringing intense rainfall that affected over 8.6 million people, displaced 745,000, and caused 300 casualties, with the Bicol Region severely impacted [3]. As flooding effects intensify, traditional early warning systems relying on water level monitoring sensors have shown critical limitations related to power, network stability, and range. Additionally, water level information alone is insufficient for timely evacuation since water levels can rise rapidly during flooding events [4].

Aside from climate change, human activities such as illegal mining, deforestation, and other harmful activities have brought negative impacts on the environment, whose effects are now widely felt. This has raised urgent concerns, especially for those living in low-lying areas, making flood detection and early warning systems a critical need for communities in order to mitigate risks and aid in preparation. Such systems can be used to mitigate risks related to flooding and enhance disaster preparedness, aligning with the United Nations Sustainable Development Goal (SDG) 13 on Climate Action. Additionally, the system aims to build safer and more resilient communities by providing information that supports overall planning within LGU disaster preparedness programs. By fostering innovation through the integration of data from different sources, this study introduces a data-driven flood monitoring and forecasting system, aligned with SDG 9 on Industry, Innovation and Infrastructure and SDG 11 on Sustainable Cities and Communities.

Flood warning systems built upon prediction models present a proactive method in hazard assessment and flood management [5], that produce accurate predictions using the volume, quality, and diversity of data rather than prior knowledge of the phenomenon [6]. Development of data-driven forecasting systems has been widely studied and supported as a means of disaster preparation. These innovations play a vital role in fostering disaster resiliency and preparedness. The purpose of this study is to develop a data-driven flood monitoring and forecasting system that visualizes data processed by an Artificial Intelligence and Machine Learning (AI/ML) engine, thereby enhancing flood risk management and community preparedness.

1.2 Review of Related Literature

Non-Data-Driven vs. Data-Driven Approach in Flood Monitoring

Flooding events drive communities to develop resilient early warning and monitoring systems with traditional approaches relying on physical technologies such as sensors, LoRaWAN and image processing. These methods are effective but constrained by high implementation costs, maintenance requirements and communication challenges [7, 8, 9, 10].

Several limitations include ultrasonic sensors becoming prone to measurement errors when it comes close to water surfaces [11] and limited battery capacity with high short circuit risk [12]. They also struggle in capturing the complex and nonlinear flood events [13] and of urban hydrological systems [14] and various unseen events also affect measurements of flood depth [15].

[16] observed that despite availability of studies applying IoT concept in flood monitoring, they fail to address the issues related to connectivity range, microcontroller processing capabilities, sensing accuracy, energy consumption and practical implementation which underscored the need for more efficient and reliable FMFS.

Data-driven approaches outperform traditional methods [17] and help create informed flood prevention decisions [18, 19] as well as enhance emergency response and preparations [20]. It addresses the gaps of traditional methods [21] demonstrating enhanced accuracy [22] and effectively simulates flood processes [23] maintaining reliable performance across all times [24].

The integration of models such as Machine Learning (ML) and Artificial Intelligence (AI) showcase capabilities in handling different tasks in FMFS [25, 26]. Advances in ML and Deep Learning (DL) redefine FMFS which brought enhanced accuracy in forecasting ability [27, 28] by reducing computational time [29, 30] and ability to handle complex data [31]. This supports [32] and [33] as both show the application of ML and AI significantly improves accuracy and computational efficiency of FMFS, and [34] demonstrates the ability of data-driven models to accurately estimate flood. Lastly, the data-driven approach is useful in monitoring rivers as shown in the results of [35] and [36] as both demonstrate its effectiveness in small to large river basins.

The adoption of a data-driven approach with the applications of AI and ML models has given significant enhancement in monitoring and forecasting flood events. This shift provides tools in assisting decision-making efforts ensuring the safety of communities and resiliency to risks particularly in flooding.

Data Integration and Modeling

Building data-driven FMFS, several data need to be collected and processed as scarcity of observed data and limitations of only one pose significant challenges [37, 38], to enhance accuracy and response time of FMFS [39, 40]. Data integration plays a vital role in a data-driven approach in FMFS as it would enhance flood prediction by having a holistic view of factors contributing to flood events.

Data integration techniques can significantly improve the detection of sudden flood events in coastal urban areas [41]. With more data available, it creates a comprehensive flood dataset [42] leveraging different data including weather forecast and urban hydrology to provide better flood prediction [43, 26]. Building on this, [44] uses feature separation combining two models to outperform other approaches in accuracy.

Data integration is essential for processing complex datasets as designs lacking temporal modeling tend to perform poorly in unseen events [45, 46]. [47] notes that between 1993 and 2020, Time Series Models (TSM) were most used while ML models from 2011 to present are becoming more complex and data-driven which enhances accuracy and dependability though costly and requires high technicality.

Big data gains prominence in showing effectiveness in forecasting and risk management [48, 49] and techniques such as connecting temporal forecasting with spatial risk assessment is used to reduce impacts of flooding [50]. Studies also suggest that Convolutional Neural Networks (CNN) model outperforms physical-based models on hidden watersheds and have minimal processing time [51, 6] while Feed Forward Neural Network (FFNN) trained using multiple regional parameters provide 84% accuracy [52]. Long Short-Term Memory (LSTM) methods also show capability in predicting streamflow [53]; combined with CNN it produces ConvLSTM (Convolutional Long Short-Term Memory), which would effectively reflect dynamics of flood stages [54].

Integrating Neural Network (NN) and Deep Learning as well as Peripheral Component Analysis (PCA) provides efficient and fast computation time [55, 30, 56]. Advancements in satellite-based remote-sensing algorithms allow global flood observation and address challenges such as cloud cover, flooded vegetation and complex urban settings [22, 57, 58]. Artificial Neural Network (ANN) is used to find relationships within data while Adaptive Neuro-Fuzzy Inference System (ANFIS) and Data Assimilation (DA) produce more accurate and timely predictions [59, 60]. AI-enhanced hydrological and ML models enhance forecast accuracy and identify vulnerable areas [61, 62].

The development of data integration and modeling techniques provides innovative and modern solutions in terms of flood monitoring and forecasting. The ability of the system to aggregate different data translating into valuable insights has proven significant in guiding decision-making, offering a new and practical way in achieving disaster resiliency. By enhancing the capability in flood prediction and data processing, it bridges the gap in traditional methods in FMFS.

Proactive Warning Systems

One of the objectives of data-driven flood monitoring and forecasting is to become proactive by leveraging different sources using AI models as they are capable of processing large data at a fast rate [63, 64]. Flood Early Warning System (FEWS) enables early warning from flood risk but remains a challenge in urban landscapes [65, 66, 35] and flash floods due to complexity [67].

AI/ML becomes a factor in how traditional FEWS shifts from reactive to proactive [68, 4] as there is a growing need for reliable flood forecasting systems due to urbanization and climate change [69, 70, 71]. Several studies [72, 73, 74] highlighted that proactive warning systems provide critical information which aids in emergency response.

The use of ML and LSTM networks shows remarkable performance in estimating workflow, identifying patterns and enhancing accuracy [75, 76, 22]. LSTM emerged as the dominant algorithm [27] among others while ML used in hybrid frameworks overcomes limitations of physical-based models [43, 77]. Deep learning models gained popularity in flood forecasting due to improved accuracy and computational ability compared to traditional ML methods [78, 20].

Several systems and models were also explored such as Extreme Gradient Boosting (XGBoost) that forecasts flood stages without requiring external triggers [79], Earth Observation (EO) based systems which show near-real-time flood depth estimates [80], Markov chains indicate improved forecast accuracy alongside existing emergency protocols [81] and community-focused EWS capabilities [82]. The hydrometeorological method is able to accurately forecast flash floods in real time [83]. The PSO–TCN–Bootstrap flood forecasting model showcases superior capability and robustness in river basins [84]. The hybrid framework of Deep Q-Network (DQN) and Twin Delayed DDPG (TD3) addresses various challenges from uncertainty [85]. Also, the use of Big Data provides accurate and reliable monitoring of water body inundation areas while effectively assessing disaster severity [86].

The evolution of data processing especially with advancements of different models paved the way in transforming from simple water level monitoring systems into more intelligent and proactive warning systems. This advancement fills the gap of traditional systems where data may be unpredicted especially in times where floods tend to rise fast. This development ensures that proper flood risk management would be implemented by providing data-driven insight. It enables better flood readiness through early warning giving ample time to prepare.

1.3 Synthesis of the State of the Art

Traditional flood monitoring and forecasting systems use physical-based technologies [7, 8, 9, 10] but limitations including high costs, maintenance and communications exist. Also, measurement errors in ultrasonic sensors nearing water surfaces [11], battery capacity and short circuit [12], difficulty capturing complex and nonlinear flood events [13], challenges in modeling urban hydrological systems [14], as well as unseen events on flood depth measurements [15]. Despite existing studies, they fail to address connectivity range, microcontroller processing capabilities, sensing accuracy, energy consumption, and practical implementation which shows the need for more efficient and reliable FMFS [16].

Data-driven approaches outperform traditional methods [17], by supporting flood prevention decisions [18, 19] that enhance emergency response [20], addressing gaps [21], enhancing accuracy [22], simulating flood processes [23], and maintaining reliable performance across varying conditions [24]. The integration of ML and AI models expanded FMFS capabilities [25, 26], with Deep Learning forecasting accuracy [27, 28] reducing computational time [29, 30] and handling large data [31], as stated by [32, 33, 34], with effectiveness that shows in small to large river basins [35, 36].

Data integration and modeling have become the foundation to scalable and accurate FMFS as scarcity of observed data and limitation of single sources pose significant challenges [37, 38]. It enhances system accuracy and response time [39, 40]. Integrating different data improves detection of sudden flood events in coastal urban areas [41], enables comprehensive flood datasets [42], and leverages weather forecasts and urban hydrology for better prediction [43, 44]. Feature separation techniques outperform existing approaches [44]. Designs lacking temporal modeling perform poorly on unseen events [45, 46]. The evolution from Time Series Models to increasingly complex ML models brought enhanced accuracy and dependability, but at higher technical cost [47].

Big Data has also proven effective in forecasting and risk management [48, 49], with techniques connecting temporal forecasting and spatial risk assessment reducing flood impacts [50]. CNN models outperform physical-based models on hidden watersheds with minimal processing time [51, 6], FFNN trained on regional parameters achieves 84% accuracy [52], and LSTM demonstrates strong streamflow prediction capability [53], combining with CNN as ConvLSTM reflects flood stage dynamics [54]. Integrating NN, DL, and PCA provides efficient computation [55, 30, 56], and satellite-based remote sensing provides global flood observation and addresses challenges such as cloud cover and urban complexity [22, 57, 58]. Models such as ANN, ANFIS, and Data Assimilation produce more accurate and timely predictions [59, 60], while AI-enhanced hydrological models identify vulnerable areas and improve forecast accuracy [61, 62].

The advancement of AI and ML paved the way for proactive FEWS from reactive [68, 4] due to growing demands [69, 70, 71]. Proactive systems offer critical information aiding emergency response [72, 73, 74]. However, they continue to face challenges particularly in urban landscapes [65, 66, 35] and flash flood scenarios due to complexity [67], which AI/ML models address through processing large data [63, 64]. LSTM networks emerged as the dominant algorithm for streamflow estimation and pattern recognition [27, 75, 76, 22] and hybrid ML frameworks overcome limitations of physics-based models [43, 77]. Deep learning models have gained prominence due to improved accuracy and computational efficiency over traditional ML methods [78, 20].

Various systems and models were explored, including XGBoost for trigger-free flood stage forecasting [79], EO-based systems for near-real-time flood depth estimation [80], Markov chains for improved forecast accuracy alongside emergency protocols [81], community-focused EWS capabilities [82], hydrometeorological methods for real-time flash flood forecasting [83], the PSO–TCN–Bootstrap model that was superior on river basins [84], a hybrid Deep Q-Network and Twin Delayed DDPG framework for addressing uncertainty [85], and Big Data-driven monitoring for accurate inundation assessment and disaster severity evaluation [86].

1.4 Gap Bridged by the Study

While various studies explored FMFS using machine learning and AI across different community needs and environments, limited research discusses the effectiveness of integrating multiple data sources for proactive monitoring at the barangay level. This study addresses this gap by developing a data-driven FMFS that provides barangay-scale forecasting, offering local disaster response units a more reliable tool for emergency preparedness. By enabling early warnings and valuable insights, the system contributes to disaster preparation efforts and promotes more resilient, informed communities in flood-prone areas.

1.5 Theoretical Framework

This study is grounded in three theories cited in [5] to guide the development and implementation. The primary theory is the DeLone and McLean (D&M) IS Success Model, which evaluates variables influencing information system success: information quality, system quality, service quality, use, user satisfaction, and net benefits. This model guides the study by ensuring high system and information quality ensuring the system is both functional and usable to its intended users.

One supporting theory is the Unified Theory of Acceptance and Use of Technology (UTAUT), which suggests that behavioral intent is the primary determinant of technology use, shaped by performance expectancy, effort expectancy, social influence, and facilitating conditions [87]. UTAUT guides the study through user-centric system design, ensuring the system meets user expectations in terms of effort and performance.

Another supporting theory is the Diffusion of Innovation (DOI) Theory, popularized by Rogers, which focuses on how innovation is accepted by users through five key factors: complexity, observability, relative advantage, trialability, and compatibility. DOI justifies the need for innovation in barangay-scale FMFS by assessing how well the proposed system addresses existing gaps and how users perceive it as an improvement over current solutions.

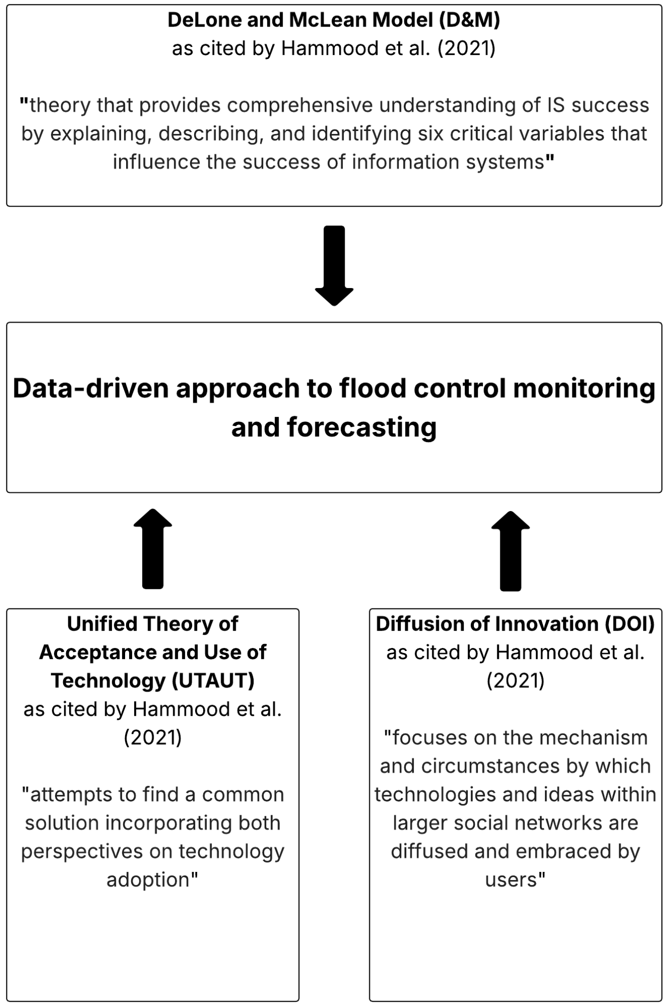

Figure 1.1: Theoretical Framework

1.6 Conceptual Framework

This study implemented the enhanced system paradigm (Input-Process-Output-Outcome). The input will come from data gathered from users and experts on what system requirements are needed, including quality indicators in developing the data-driven approach in flood monitoring and forecasting. These will serve as the guide in designing the system prototype which integrates AI and machine learning. A series of functional testing will be conducted to ensure that the prototype adheres to the user's requirements. The final output of the study is the prototype that will provide the data-driven system through the AI/ML module integrated. This is expected to enhance the decision-making activities of the barangay officials to ensure the safety of the community during flood season.

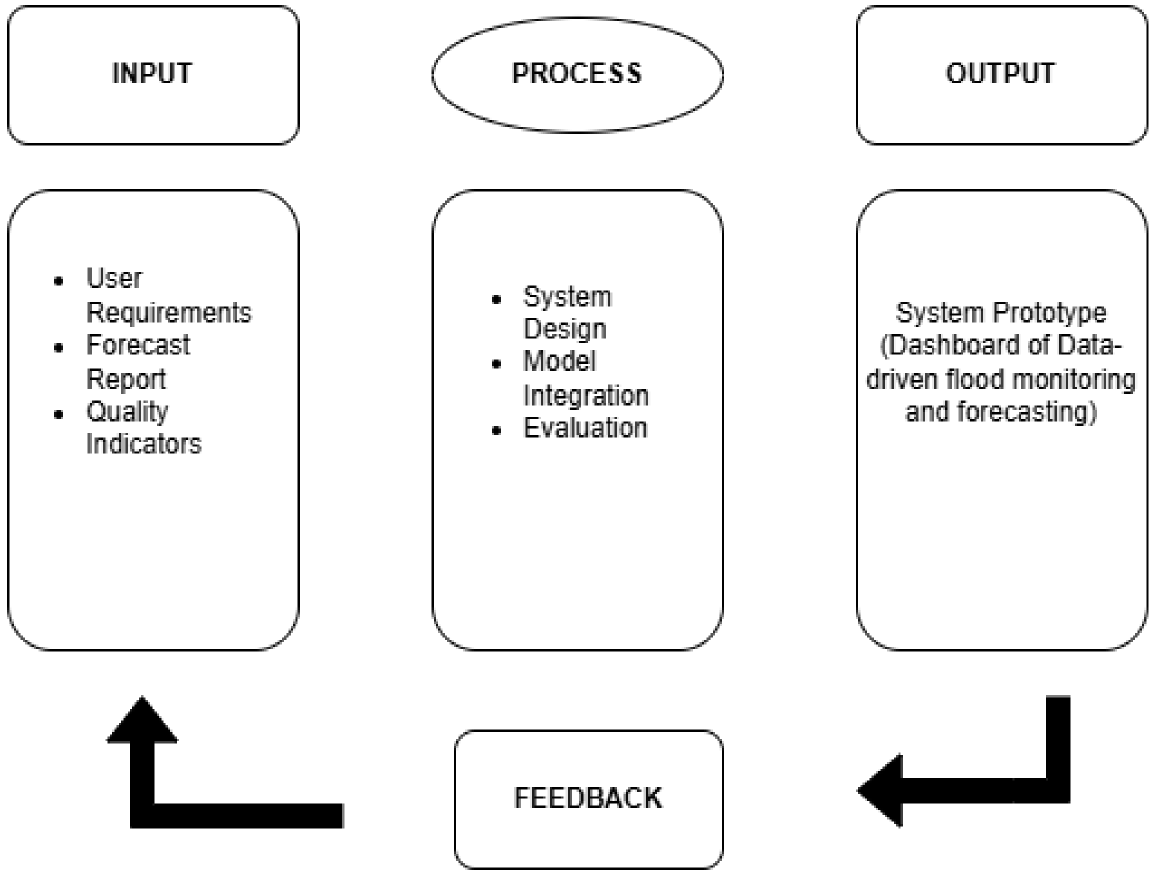

Figure 1.2: Conceptual Framework

1.7 Statement of the Problem

The main problem of the study is to design and develop a system that aims to fill the gaps in using traditional sensor-based flood monitoring. For successful development and implementation of the system, the following questions would be used as a guide:

1. What is the current status of disaster management in monitoring flooding in the barangay?

2. What are the specific limitations and challenges encountered by the disaster management team using current flood monitoring methods?

3. What are the critical functional and non-functional requirements needed to develop a proactive, data-driven flood monitoring and forecasting system tailored to the needs of the barangay?

4. What is the level of compatibility between the proposed system and the existing operational procedures and resource capabilities of the disaster management team?

5. What visualization methods for flood forecasting data can ensure easy interpretation and actionability for non-technical LGU officials during emergency situations?

6. What is the level of user acceptability of the developed system among officials and residents in terms of usability, reliability, and functionality?

1.8 Assumptions

The following assumptions were made:

1. The current traditional or sensor-based flood monitoring methods employed have inherent limitations, latencies, or inefficiencies that hinder optimal disaster response and decision-making.

2. It is assumed that the disaster management team faces operational, technical, or resource-related challenges with their existing flood monitoring methods that warrant investigation and potential improvement.

3. There are specific, identifiable functional and non-functional requirements that, when integrated into a data-driven system, will effectively address the gaps in the barangay's current flood monitoring capabilities.

4. The proposed data-driven system can be designed to be compatible and interoperable with the existing operational procedures, technical resources, and workflow of the disaster management team.

5. Effective data visualization techniques significantly improve the ability of non-technical LGU officials to interpret complex forecasting data and make faster, more accurate decisions during emergencies compared to raw data presentation.

6. Officials and residents are willing to adopt new technology and that their level of acceptance can be accurately measured through indicators of usability, reliability, and functionality.

1.9 Scope and Delimitation

The proposed Data-driven Flood Monitoring and Forecasting System aims to assist stakeholders in creating educated decisions. At the end, the study aims to deliver a functional application capable of visualizing data from the built model.

The research would focus on development and evaluation of a visual dashboard and alert notification for a data-driven FMFS. This covers integration of the forecast data outputs that would come from an AI/ML model. The interface presents the forecast data. The system development will follow an Agile methodology to allow continuous feedback, refinement, and adaptation throughout the development process.

System design would prioritize simplicity in User Interface (UI) layout for better User Experience (UX).

To realize the success of the project, evaluation would be conducted such as functionality and quality tests using survey-based evaluation. It would also involve participation from barangay DRRMO and selected residents as system evaluators and pilot testers. Statistical analysis includes percentage, mean/weighted mean, and Likert-based interpretation for evaluating user perception. Survey instruments are provided in English and Bikol dialect to ensure user comprehension with the system interface utilizing Bikol and English language.

The study is limited to be conducted in Barangay Triangulo, Naga City. This would utilize information implemented from their operational procedures with respect to the data needed in the design and development of the dashboard system. No physical sensor development would be involved including installation. All data would come from external sources and government agencies only.

1.10 Significance of the Study

The study sought to benefit the following stakeholders.

Residents. The system can empower residents by providing them a more reliable and accurate flood prediction user-centric system.

Barangay LGU. The data visualization, real-time updates and proactive warning system allows informed decision-making from the disaster management team especially in flood risk mitigation.

City Government. The city government can also benefit from the data gathered for flood mitigation and response. It also aligns with the government's aims for innovation especially in digital technology.

Educational Institutions. Schools within the area can benefit from the system's early warning capabilities to make informed decisions regarding class suspensions, student safety protocols, and campus evacuation procedures.

Other Researchers. The results of this study can be used as a reference for future studies in related fields. The identified requirements can also serve as guidelines in building similar monitoring tools.

1.11 Definition of Terms

This section aims to define terms used in the context of the research for better understanding.

Data-Driven Approach. The use of AI/ML models to analyze large datasets for forecasting, particularly useful in areas with limited physical measurements [36]. In this study, this refers to the use of different data sources to forecast flood events to create insights that support decision-making.

Flood Control Monitoring. In [7], it is the use of various sensors and processes to capture flood events. In this study, it refers to the continuous observation of meteorological data related to flooding conditions.

Flood Control Forecasting. The use of ML models to create a timely and accurate warning to reduce flood-related risks for future predictions [75]. In this study, it refers to the prediction of potential flood events based on computed models, trends, or real-time data inputs.

Decision Makers. Persons primarily responsible for making informed decisions [74]. In this study, decision makers refer to individuals or groups responsible for responding to flood information, such as LGU officials, emergency managers, barangay personnel, and planning units.

Community Resilience. The ability of a community to adapt and respond to disaster or harmful situations. In this study, community resilience refers to the capacity of local communities to prepare, respond, and recover from flood impacts, represented through preparedness levels and improved decision-making enabled by the system.

Early Warning System (EWS). An integrated system used to monitor flood conditions and provide warnings for risk mitigation and preparation [5]. In this study, an early warning system refers to mechanisms that detect flood risk and issue alerts to decision makers or the community, represented through notifications, alerts, and automated warning triggers.

User Interface (UI). The visual and interactive components through which users access and interact with flood monitoring systems, including dashboards, maps, and alert displays [11]. In this study, UI refers to the dashboard layout, design elements, navigation components, and visual presentation of flood forecasting data that enable users to access and interpret information efficiently.

User Experience (UX). Accessible and understandable presentation of flood information to stakeholders [10]. In this study, UX refers to the measured usability, user satisfaction, and effectiveness of the system interface as evaluated through user feedback and response times during system interaction.

Real-Time Data. Information that is delivered immediately after collection with minimal delay [11]. In this study, real-time data refers to information that is continuously updated and processed by the system enabling current situational awareness for flood monitoring.

Disaster Preparedness. Proactive measures, plans, and systems put in place before a flood event to minimize impacts [18, 5]. In this study, disaster preparedness refers to the actions taken by LGU officials and residents based on system forecasts, including evacuation planning, resource allocation, and positioning of emergency response teams.

System Integration. Combining multiple data sources into a single dataset [2]. In this study, system integration refers to the technical implementation that connects AI/ML forecast models with data sources, dashboard interface, and alert notification mechanisms to create a cohesive flood monitoring and forecasting platform.

2 Methodology

This section presented the methods and techniques used and employed. This includes the data required, data gathering methods, population, research instruments as well as system architecture and design applied.

2.1 Research Design

This study used a descriptive research design to describe the current flood monitoring practices and identify user requirements for a data-driven flood monitoring and warning system. A mixed-method approach was used to enable a comprehensive analysis and assessment of system performance, user perception and adoption factors. The usage of mixed methods allows the study to leverage both qualitative and quantitative offering a richer interpretation and validation.

Agile methodology would be used for software development allowing better collaboration, engagement and continuous refinement within the whole development process. Development sprints and prototype releases would be employed to gather feedback to allow problem identification and mitigation as early as the development phase, minimizing errors, bugs and unnecessary changes after deployment.

2.2 Respondents and Locale of the Study

The study will be conducted in Barangay Triangulo, Naga City, Camarines Sur, one of the identified flood-prone areas within the city. Due to its location, residents often experience flooding that disrupts daily activities and poses safety risks when heavy rains occur. Flooding in the area tends to escalate quickly, and many households depend on announcements from barangay officials or social media updates, which sometimes leads to delays in receiving important information. However, because the current system uses manual communication methods, real-time updates are sometimes slow or inconsistent.

Given the challenges, Barangay Triangulo is an appropriate site for the development and testing of the proposed Data-driven Flood Monitoring and Early Warning System. This allows the researchers to understand the actual experiences of residents who encounter flooding regularly and to gather insights from local officials handling disaster operations. This helps ensure that the proposed system is practical, responsive, and aligned with the real needs of the community.

Sample Size Determination

To ensure adequate representation of the target population, the sample size was computed using Yamane's Formula, which is appropriate when the total population is known but its variability is uncertain.

$$n = \frac{N}{1 + Ne^2} \tag{2.1}$$

where:

- n = required sample size
- N = total population of Barangay Triangulo
- e = margin of error (commonly set at 0.05)

2.3 Data Gathering Procedure

To ensure a comprehensive collection of data, the researchers employed multiple data gathering methods. These methods were selected to obtain both quantitative and qualitative information relevant to the study, ensuring that the findings accurately reflect the current state and needs of the selected barangay.

Surveys. Comprises a series of questions often with selections and made to gather information from the respondents. It would include open-ended questions to enable the respondents to answer freely.

Document Analysis. A supporting method that aims to gain understanding about the current methods and systems being done within the selected barangay. This provides understanding about possible improvements that can be made.

2.4 Research Instruments

This study would employ different research instruments to gather data such as a Survey Questionnaire and Evaluation Tool. Each instrument was designed to address specific research objectives and ensure the validity and reliability of the data collected.

Survey Questionnaire. A set of structured questions with provided selections that aims to understand the population as a whole.

Evaluation Tool. Used to assess the system design and performance. It helps the researcher analyze feedback and data gathered that helps make the system usable and accessible.

2.5 Ethical Considerations

This study strictly adheres to ethical standards to ensure the protection of participants and the responsible handling of data. The following considerations were observed:

Informed Consent. Participants involved were fully informed about the purpose, procedures, and scope of the study. They were provided with clear instructions and had the freedom to voluntarily participate or withdraw at any time without negative consequence. Written consent was obtained to ensure acknowledgment and agreement.

Data Privacy and Confidentiality. All personal information and data collected from participants were treated with strict confidentiality. No identifiable data would be collected. Access to raw data were limited to the research team only.

Fair Use of Data. Data collected were used solely for the purposes of this research. Any reporting of findings ensured that individual participants could not be identified.

Minimization of Risk. Measures were taken to minimize any potential physical, psychological, or social risks to participants.

Compliance with Legal and Institutional Guidelines. This study complied with relevant laws, institutional policies, and ethical standards concerning research involving human subjects, data collection, and environmental monitoring.

2.6 Statistical Tools

The study will make use of the following statistical tools to analyze and interpret the data collected from the research instruments.

Percentage Distribution. Percentage was used to describe respondent characteristics and summarize categorical responses such as demographics, flood awareness, or technology adoption readiness. The result will be calculated using the formula:

$$P = \frac{f}{N} \times 100 \tag{2.2}$$

where:

- P = Percentage
- f = Frequency of responses
- N = Total number of respondents

Mean / Average. The mean was used to determine the central tendency of responses where all values are equally weighted. It will be computed using the formula:

$$\bar{X} = \frac{\sum x}{N} \tag{2.3}$$

where:

- $\bar{X}$ = Mean

- $\sum x$ = Sum of all values
- N = Number of responses

Weighted Mean. Weighted mean was applied to interpret Likert-scale responses, recognizing that each response level carries different significance. The result would be determined using the following formula:

$$WM = \frac{\sum(w \times X)}{\sum w} \tag{2.4}$$

where:

- WM = Weighted Mean
- w = Assigned weight (e.g., 1–5)
- X = Frequency of responses

2.7 Statistical Treatment of Data

The study will interpret the data gathered using a 5-point Likert scale. This was used to gather respondents' perception regarding the quality, functionality and user acceptance of the system.

Likert Scale. In this study, a 5-point Likert scale would be used to measure the respondents' perceptions of the system in terms of quality, usability, reliability, information value, and acceptance with 1 being the lowest and 5 being the highest to determine the overall rating.

Table 2.1: Likert Scale for System Quality, Information Quality and Usability

Range	Verbal Interpretation (System Quality)	Verbal Interpretation (Information Quality)	Verbal Interpretation (Usability)
4.21 – 5.00	Excellent	Very High	Very Highly Usable
3.41 – 4.20	Good	High	Highly Usable
2.61 – 3.40	Neutral	Moderate	Moderately Usable
1.81 – 2.60	Poor	Low	Low Usability
1.00 – 1.80	Very Poor	Very Low	Very Low Usability

This outlines the Likert scale ranges and their corresponding interpretations for evaluating System Quality, Information Quality, and Usability, where higher scores indicate more favorable perceptions of system performance, information accuracy, and overall user usability. This scale enables interpretation of user responses by categorizing results into levels that reflect the degree of effectiveness, reliability, and usability of the system.

Table 2.2: Likert Scale for Reliability, Functionality and User Acceptance

Range	Verbal Interpretation (Reliability)	Verbal Interpretation (Functionality)	Verbal Interpretation (User Acceptance)
4.21 – 5.00	Very Highly Reliable	Very Functional	Very Highly Acceptable
3.41 – 4.20	Highly Reliable	Functional	Highly Acceptable
2.61 – 3.40	Moderately Reliable	Moderately Functional	Moderately Acceptable
1.81 – 2.60	Low Reliability	Less Functional	Low Acceptable
1.00 – 1.80	Very Low Reliability	Not Functional	Very Low Acceptable

Table 2.2 presents the Likert scale ranges for assessing Reliability, Functionality, and User Acceptance, where higher scores indicate more favorable evaluations. This scale categorizes user responses to measure system effectiveness, consistency, acceptability, and user adoption willingness.

2.8 Systems Design

This section highlights various designs for the proposed data-driven flood monitoring and forecasting system. This includes system architecture, use case diagram and system flow.

Use Case. The use case design of the proposed FMFS shows the interaction between the system, users and external data providers. This highlights how each actor (residents, barangay DRRMO, PAGASA and third-party entities) uses the system and helps increase clarity on how the system works.

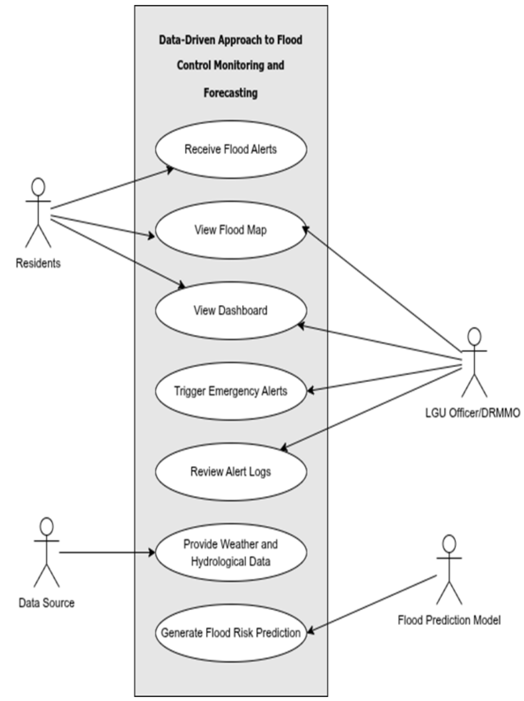

Figure 2.1: Use Case Diagram

Data Flow Diagram. This illustrates the Data Flow Diagram (DFD) which illustrates how data flows within the system.

The first figure is the Level 0 of the proposed system. This provides a high-level overview of the system and interaction with all external entities. The system revolves around four external entities — the Admin, Data Sources, Barangay DRRMO/Officials, and Residents — each playing a distinct role throughout the system. The admin handles the registration for barangay officials and the officials handles the residents registration as well. The system uses data sources which is essential for flood monitoring and forecasting. All forecast reports are sent to barangay officials and residents whereas the officials are able to send emergency alerts to the residents. Processed information are stored in data storage for retrieval and reference.

Level 1 of DFD breaks down the system through five processes: user registration, data collection and processing, forecast generation, alert dissemination, and report generation. Processed data is used to generate forecasts which are then distributed to residents and DRRMO officials, while real-time emergency alerts are issued upon detection of critical flood conditions. All alerts and system activities are compiled into historical reports, supporting post-event analysis and long-term disaster preparedness planning.

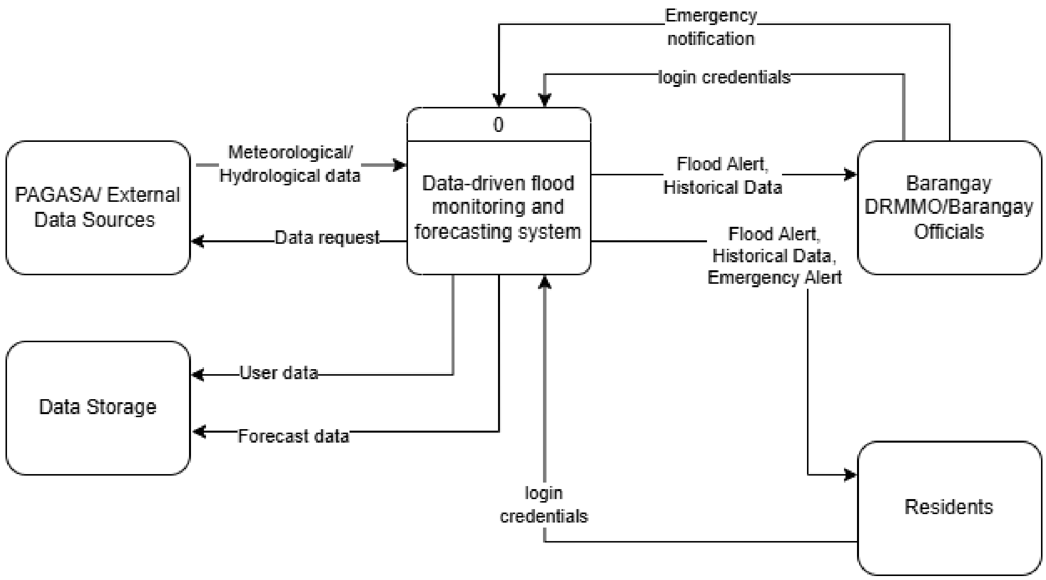

Figure 2.2: Proposed System Context Diagram

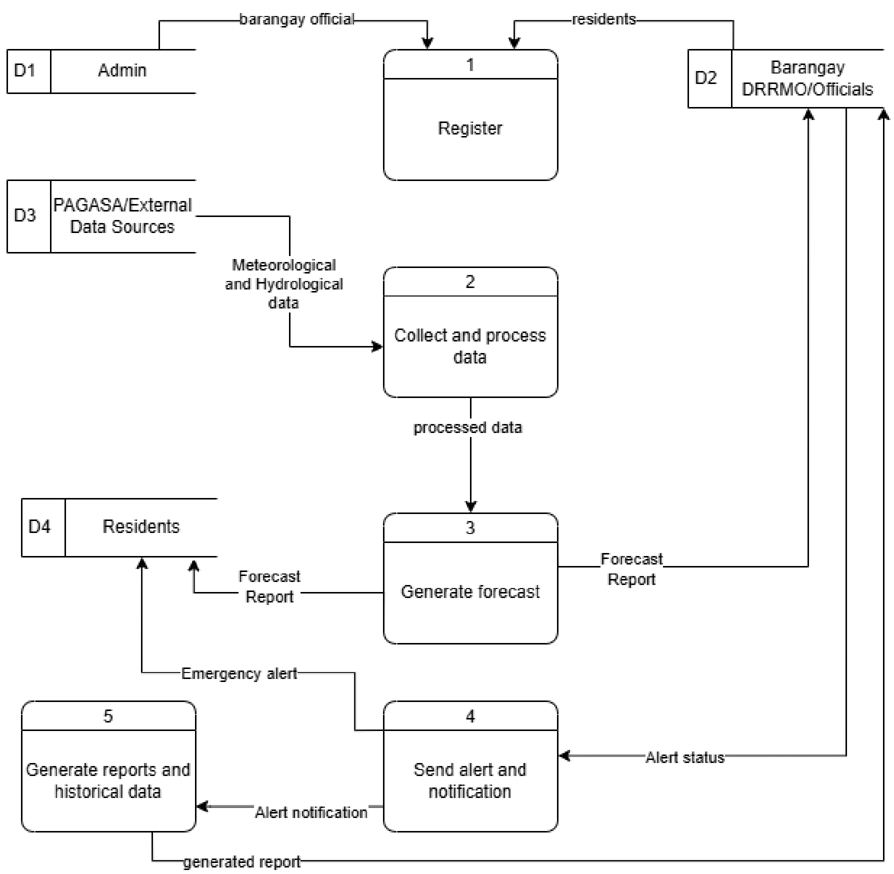

Figure 2.3: Proposed System Level 1 Diagram

Entity-Relationship Diagram (ERD) The ERD illustrate the entities that are used within the system. This also shows their connection and individual attributes.

This serves as blueprint for the design of database during system development which ensures that data relationships are well defined and aligns with functional requirements of the system. It also serves as a communication tool betwween technical and non-technical team members, making the system's structure easier to understand and dicuss.

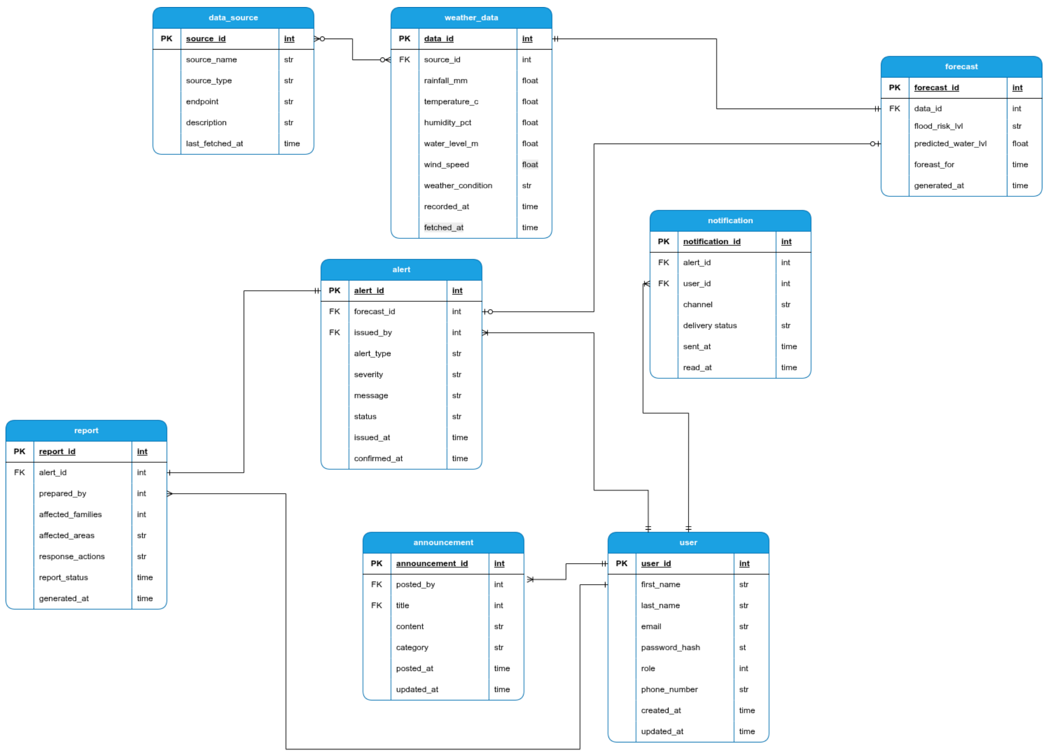

Figure 2.4: Entity Relationship Diagram

System Flow. The Business Process Model and Notation (BPMN) for the Proposed Data-driven FMFS is divided into five distinct pools representing the system's modular architecture: Data Providers, Data Ingestion, AI/ML Forecasting, Alert Engine, and the Dashboard interface.

The process initiates in the Data Providers pool, where external weather and hydrological data are released. The Data Ingestion Module operates on an automated timer event, triggering a fetch request to the PAGASA API every five minutes. Upon retrieval, the system enters a validation loop; a decision gateway checks if the data is complete. If the data is incomplete, the system logs the error and requests missing data. If the data is valid, it is stored in the database and forwards the clean data. The AI/ML Forecasting Engine receives this clean dataset to execute the prediction model. The engine processes the variables to generate a specific Flood Risk Level, which serves as the input for the alerting logic. The Alert Engine acts as the central decision node.

Upon receiving the risk level, an exclusive gateway routes the process into three distinct workflows based on severity—High Risk, where the system triggers emergency alerts and sends immediate push notifications to residents; Moderate Risk, where the system sends an early warning advisory; and Low Risk, where the system only updates the internal system dashboard and logs the update. The process concludes in the Dashboard pool where the status update and forecast data would be reflected and show the relevant advisories to the end-users.

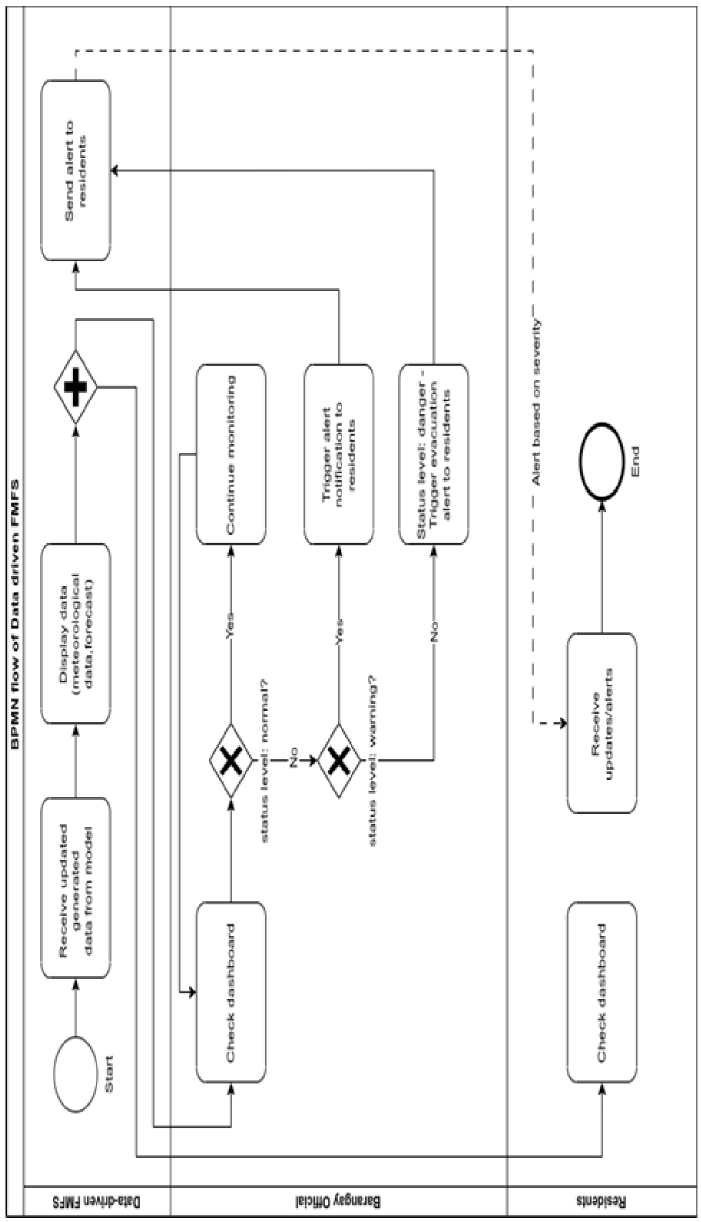

Figure 2.5: Business Process Model Notation for Proposed Data-Driven Flood Monitoring and Forecasting System

System Architecture

The overall system architecture is designed to support a data-driven flood monitoring and early-warning platform that integrates external hydrological data sources, an AI/ML prediction engine, and end-users such as residents and Barangay DRRMO personnel. The diagram illustrates the flow of data and information among the major components of the system.

Hydrological and meteorological data are obtained from PAGASA and other third-party data providers, which serve as the primary external data sources. These datasets, which may include rainfall intensity, river gauge levels, soil moisture, and weather forecasts, are transmitted to the AI/ML Engine for processing. The AI/ML Engine performs predictive analytics based on the incoming hydrological data. It applies trained models to generate flood predictions, severity estimates, and early-warning insights. The outputs of the AI/ML engine are then forwarded to the Server, which acts as the central system controller. The Server receives the predicted flood events or system update and produces visual flood forecasts, which are made accessible to Barangay DRRMO users through their desktop or laptop interfaces. These users can view system-generated alerts, monitor predictions, and manage local response actions through the dashboard. When the system detects potentially hazardous conditions, the server sends emergency warnings directly to residents through their mobile devices. This enables rapid and proactive distribution of alerts to the community.

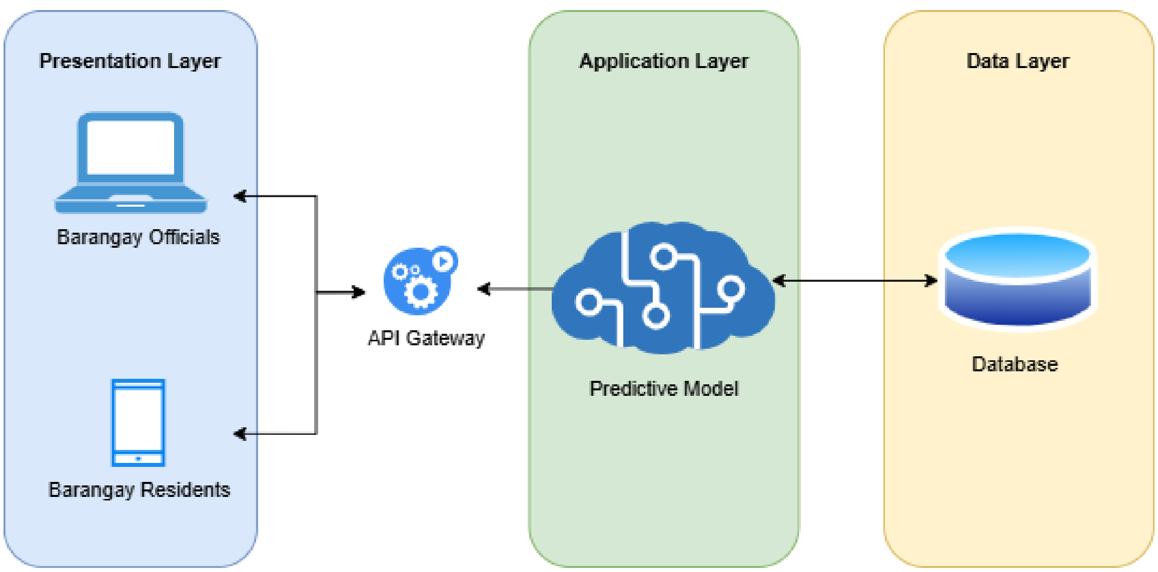

Figure 2.6: System Architecture of Proposed Data-driven Flood Monitoring and Forecasting System

Project Timeline

The project execution is structured over a twelve-month timeline, segmented into six key phases: Initiation, Design, Development, Testing, Implementation, and Maintenance.

The project commences in Month 1 with Project Initiation, focusing on Requirements Gathering and Technical Research to establish the system's full scope and objectives. These activities extend into Month 2, allowing the team to thoroughly document user needs and evaluate relevant technologies before formal design work begins. The Design phase runs concurrently from Month 2 to Month 3, during which the team defines the System Architecture, finalizes the Database Design, and creates UI/UX Wireframes to guide subsequent development efforts.

Development spans Months 3 through 8, beginning with the Data Integration Module and Data Cleaning and Storage mechanisms, followed by the Backend API Setup and initial Dashboard Development in Months 4 through 6. The latter half of Development—Months 6 through 8—focuses on the integration of the AI/ML engine, Alert Rule Setup, Alert System implementation, and full System Integration. From Months 8 to 10, the Testing phase is conducted, covering Internal Testing, User Acceptance Testing, and Optimization and Bug Fixing to ensure system reliability and usability before launch. Months 10 and 11 are dedicated to Implementation, during which the system is formally deployed and made operational for end-users. The final phase, Maintenance, spans Months 11 and 12 and involves documentation, capture of lessons learned, and ongoing system monitoring to support long-term quality assurance and continuous improvement.

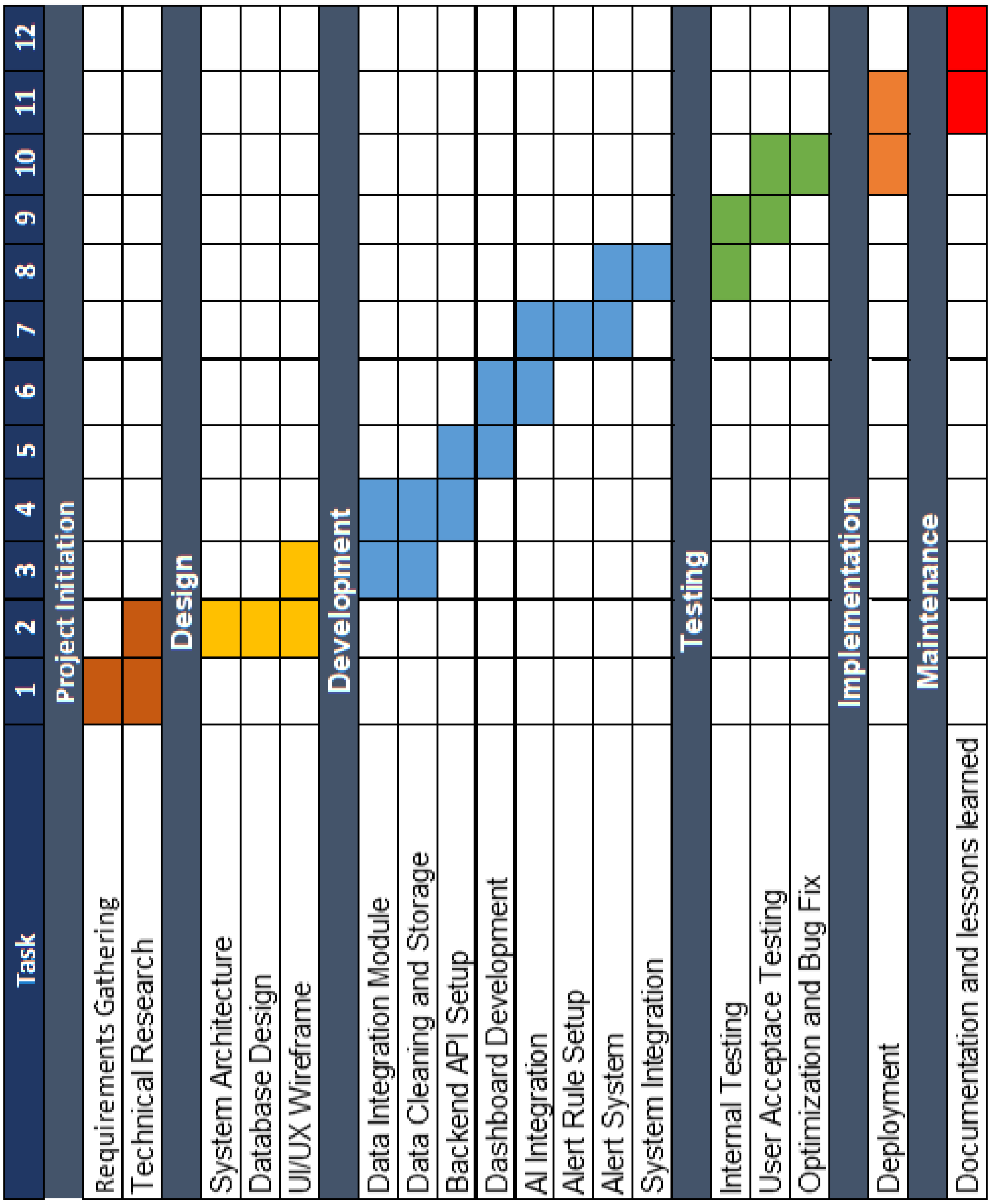

Figure 2.7: Project Timeline for Proposed Data-driven FMFS

3 Results and Discussion

This chapter presents the results of the survey conducted among barangay officials and residents from Barangay Trianggulo. The data collected were analyzed and interpreted to determine the current state of flood monitoring in the barangay, identify the key challenges faced by residents and barangay officials, and establish the system requirements and features that would best address their needs.

3.1 Profile of the Respondents

A total of 78 respondents participated in the survey. Table 3.1 presents the demographic profile of the respondents in terms of role/designation, sex, age range, educational attainment, and years of residence or service in Barangay Trianggulo.

Table 3.1: Distribution of Respondents by Role/Designation

Category	**Role / Designation**	**Freq. (f)**	**Pct. (%)**
Barangay Residents	Barangay Resident	66	84.61
Barangay Officials	Tanod / Barangay Responder	4	5.13
	Barangay Councilor (Kagawad)	4	5.13
	Brgy. Clerk	1	1.28
	Brgy. Treasurer	1	1.28
	Not specified	2	2.56
Total		**78**	**100.00**

The respondents were categorized into two groups: barangay residents and barangay officials. The majority were barangay residents ($n = 66$, 84.61%). Barangay officials comprised 10 respondents (12.82%), broken down into Tanod/Barangay Responders ($n = 4$, 5.13%), Barangay Councilors or Kagawad ($n = 4$, 5.13%), Brgy. Clerk ($n = 1$, 1.28%), and Brgy. Treasurer ($n = 1$, 1.28%). This distribution ensures that the evaluation captures both the community end-users of the system and the frontline personnel directly involved in disaster risk reduction and management (DRRM) operations.

Table 3.2: Distribution of Respondents by Sex

Sex	**Frequency (f)**	**Percentage (%)**
Male (Lalaki)	36	46.15
Female (Babae)	33	42.31
Prefer not to say	8	10.26
Not specified	1	1.28
Total	**78**	**100.00**

In terms of sex, male respondents slightly outnumbered female respondents with 36 (46.15%) and 33 (42.31%) respectively. Eight respondents (10.26%) preferred not to disclose their sex.

Table 3.3: Distribution of Respondents by Age Range

Age Range	Frequency (f)	Percentage (%)
18–25 years old	37	47.44
26–35 years old	22	28.21
36–45 years old	10	12.82
46–55 years old	6	7.69
56 years old and above	2	2.56
Not specified	1	1.28
Total	**78**	**100.00**

Table 3.3 reveals that the majority of the respondents belonged to the 18–25 years old age group (n = 37, 47.44%), followed by those aged 26–35 years old (n = 22, 28.21%). Respondents aged 36–45 accounted for 12.82%, while those aged 46–55 represented 7.69%, and those aged 56 and above comprised only 2.56%. The predominance of younger respondents (ages 18–35) comprising 75.65% of the total sample suggests a tech-savvy population that may be more receptive to technology-based solutions for flood monitoring and early warning.

3.2 Current State of Flood Monitoring

3.2.1 Current Flood Monitoring Methods

Table 3.4: Current Flood Monitoring Methods Used in the Barangay (Multi-Select)

Monitoring Method	Freq. (f)	Pct. (%)
Manual visual inspection / physical patrols	57	73.08
Social media monitoring (Facebook)	43	55.13
River water level observation — manual	36	46.15
Rain gauge observation	32	41.03
Reports from residents via phone or text	32	41.03
Coordination with PAGASA / NDRRMC advisories	29	37.18
Automated sensors or IoT devices	6	7.69
None / No formal monitoring system	4	5.13

Table 3.4 reveals that manual visual inspection remains the most prevalent method, accounting for 57 respondents (73.08%), followed by social media monitoring with 43 respondents (55.13%), while manual river water level observation was reported by 36 respondents (46.15%). Rain gauge observation and resident-reported information via phone or text were each noted by 32 respondents (41.03%). Next is coordination with PAGASA and NDRRMC advisories with 29 respondents (37.18%). Six respondents (7.69%) indicated the use of automated sensors or IoT devices, while four respondents (5.13%) reported the absence of any formal monitoring system.

These highlight the barangay's reliance on manual and informal monitoring approaches. These underscores the need for a more systematic and technology-based flood monitoring and forecasting system.

3.2.2 Frequency of Flood Monitoring Activities

Table 3.5: Frequency of Flood Monitoring Activities

Monitoring Frequency	**Freq. (f)**	**Pct. (%)**
Only when typhoon / heavy rain warnings are issued	32	41.03
Only during rainy season	26	33.33
Real-time / 24/7 monitoring	9	11.54
Weekly	5	6.41
No regular schedule	4	5.13
Daily	1	1.28
Not specified	1	1.28
Total	**78**	**100.00**

Most respondents stated that flood monitoring in the barangay is conducted only when typhoon or heavy rain warnings are issued ($n = 32$, 41.03%), while 33.33% indicated monitoring occurs only during the rainy season. Only 11.54% reported real-time or 24/7 monitoring. These confirms the reactive rather than proactive nature of the barangay's current flood monitoring practices, where monitoring activities is a response to upcoming weather events rather than preventive measure.

3.2.3 Status of Flood Early Warning System (FEWS)

Table 3.6: Status of Flood Early Warning System (FEWS)

FEWS Status	**Freq. (f)**	**Pct. (%)**
Yes — but not fully operational	33	42.31
Yes — fully operational	30	38.46
No (Wala)	11	14.10
Under development	3	3.85
Not specified	1	1.28
Total	**78**	**100.00**

The table shows the respondents' awareness of the status of FEWS in the barangay. While 30 respondents (38.46%) believed that it is fully operational, majority of respondents (42.31%) indicated that although it exists, it is not fully operational. 11 respondents (14.10%) stated that no FEWS exists, and three (3.85%) reported that one is currently under development. These findings indicate that the barangay's flood early warning capabilities is insufficient. This gap shows the urgency of developing a comprehensive and functional FMFS.

3.2.4 Communication Methods for Flood Warnings

Table 3.7 summarizes the communication channels used to relay flood warnings to residents. Megaphone and public announcements were the most widely used method ($n = 53$, 67.95%), followed by Social media ($n = 46$, 58.97%). Next is SMS with 31 respondents (39.74%), while house-to-house notification by tanods was reported by 26

Table 3.7: Communication Methods for Flood Warnings (Multi-Select)

Communication Method	**Freq. (f)**	**Pct. (%)**
Megaphone / public announcement	53	67.95
Social media (Facebook group/page)	46	58.97
Text message (SMS blast)	31	39.74
House-to-house notification by tanods	26	33.33
Barangay radio	21	26.92
Community alarm / bell (kampana)	6	7.69
None (Wala)	3	3.85

respondents (33.33%). Barangay radio was mentioned by 21 respondents (26.92%). These results reveal a dependence on traditional communication methods. This raises concerns about the speed and reach of warnings, especially during nighttime or severe weather conditions.

3.2.5 Budget Allocation for Flood Monitoring

Table 3.8: Budget Allocation for Flood Monitoring Systems

Budget Status	**Freq. (f)**	**Pct. (%)**
Yes — but insufficient	40	51.28
Yes — adequate budget	25	32.05
I am not aware / unsure	11	14.10
No dedicated budget	2	2.56
Total	**78**	**100.00**

Table 3.8 presents the respondents awareness of budget allocation for flood monitoring in barangay. Majority of respondents ($n = 40$, 51.28%) indicated that while a budget for flood monitoring exists, it is insufficient. Only 25 respondents (32.05%) reported having an adequate budget, and 11 respondents (14.10%) were not aware of the barangay's budget allocation. Two respondents (2.56%) stated there is no dedicated budget at all. These results confirm that financial constraints are a significant barrier to improving the barangay's flood monitoring infrastructure.

3.3 Challenges and Limitations in Current Flood Monitoring

Respondents were asked to rate twelve (12) identified challenges using a 4-point Likert scale ranging from 1 (Strongly Disagree) to 4 (Strongly Agree). The responses were analyzed using weighted mean scores and interpreted as follows: 1.00–1.75 = Strongly Disagree; 1.76–2.50 = Disagree; 2.51–3.25 = Agree; 3.26–4.00 = Strongly Agree.

Table 3.9: Challenges and Limitations in Current Flood Monitoring

Challenge	**Wtd. Mean**	**Interpretation**
Budget constraints significantly limit flood monitoring improvements	3.11	Agree
Flood monitoring is highly dependent on manual human effort	3.04	Agree
Existing monitoring equipment is outdated or inadequate	2.87	Agree
Communication breakdowns occur when relaying flood information	2.83	Agree
Current flood monitoring methods are slow / provide no timely warnings	2.83	Agree
The barangay lacks sufficient manpower for effective flood monitoring	2.71	Agree
Flooding is often detected too late for adequate evacuation preparation	2.71	Agree
The barangay does not have access to real-time weather/water level data	2.66	Agree
Insufficient technical training among BDRRM personnel	2.64	Agree
No standardized procedures or protocols for flood monitoring	2.64	Agree
Data collected through current methods is unreliable or inaccurate	2.57	Agree
Difficulty coordinating with PAGASA and NDRRMC	2.55	Agree

Table 3.9 presents how respondents agreed with identified challenges. Budget constraints emerged as the most pressing challenge (Mean=3.11, 80%), followed by over-reliance on manual human effort (Mean=3.04, 75%) and outdated equipment (Mean=2.87, 71%). Communication breakdowns and slow warning dissemination were also rated highly at 69% and 65% respectively.

Notably, a respondent who identified as a Barangay Councilor remarked: “The technology is only 30% of the Flood System. The other 70% is maintenance, people, local knowledge and trust.” This perspective highlights that while technology is vital, institutional and human dimensions must also be addressed for any proposed system to be truly effective. Furthermore, a resident noted that drainage system problems contribute significantly to flash flooding, suggesting that the monitoring system should account for infrastructure-related flood triggers beyond rainfall and river water levels alone.

3.4 Desired Features of the Proposed System

Respondents were asked to rate the importance of twelve (12) proposed system features using a 5-point Likert scale. Results were interpreted as follows: 1.00–1.80 = Not Important; 1.81–2.60 = Slightly Important; 2.61–3.40 = Moderately Important; 3.41–4.20 = Very Important; 4.21–5.00 = Extremely Important.

Table 3.10: Importance Ratings of Desired System Features

Desired Feature	Wtd. Mean	Interpretation
Automated flood alerts / notifications	4.52	Extremely Important
Real-time water level monitoring (automated sensors)	4.52	Extremely Important
Real-time rainfall data collection and display	4.47	Extremely Important
Integration with PAGASA weather data and advisories	4.47	Extremely Important
Flood risk zone mapping (Barangay Trianggulo-specific)	4.44	Extremely Important
Dashboard showing live flood status	4.42	Extremely Important
Coordination/communication module for response teams	4.38	Extremely Important
Evacuation route mapping and guidance	4.30	Extremely Important
Flood level forecasting (hours in advance)	4.28	Extremely Important
Historical flood data recording for trend analysis	4.28	Extremely Important
Incident reporting module for BDRRM team	4.28	Extremely Important
Automated escalation to higher authorities	4.28	Extremely Important

All twelve proposed features were rated in the 'Extremely Important' range (4.21–5.00), reflecting a strong demand across all functionalities. Automated flood alerts and real-time water level monitoring ranked highest with a weighted mean of 4.52, underscoring the respondents' prioritization of real-time, automated information delivery. Real-time rainfall data and PAGASA integration followed closely at 4.47, confirming the need for reliable, agency-sourced weather information. Flood risk zone mapping (4.44) and live dashboards (4.42) also ranked highly, indicating that spatial visualization of flood risks is equally valued.

These results affirm that the proposed system must prioritize automation, real-time data integration, and accessible visual displays. The uniformly high importance ratings across all twelve features also suggest that the system should be designed as a comprehensive platform rather than a single-purpose tool, addressing the full spectrum of flood monitoring, forecasting, communication, and response needs of the barangay.

3.5 Non-Functional Requirements of the Proposed System

Respondents also rated ten (10) non-functional requirements using the same 5-point Likert scale. Non-functional requirements pertain to system quality attributes such as performance, accessibility, reliability, and sustainability.

Table 3.11: Importance Ratings of Non-Functional System Requirements

Non-Functional Requirement	**Wtd. Mean**	**Interpretation**
Must operate even during power interruptions (backup power)	4.53	Extremely Important
Must generate automated reports for DRRM compliance	4.53	Extremely Important
Must be affordable to operate and maintain	4.51	Extremely Important
Must respond and process data within seconds (low latency)	4.43	Extremely Important
Data stored must be secure and protected	4.43	Extremely Important
Interface must be simple enough for non-technical users	4.43	Extremely Important
Must be scalable to accommodate future enhancements	4.42	Extremely Important
Must be available and operational 24/7	4.41	Extremely Important
Must be accessible via mobile phone (Android/iOS)	4.38	Extremely Important
Must work with low or intermittent internet connectivity	4.28	Extremely Important

Table 3.11 shows that all ten non-functional requirements were rated within the 'Extremely Important' range. The two highest-rated requirements — backup power operation and automated DRRM report generation — both achieved a weighted mean of 4.53. Affordability of operation and maintenance (4.51) and low latency data processing (4.43) followed closely. These findings reflect the operational and infrastructural realities of the barangay, where power interruptions are a common

occurrence during typhoons and where limited budgets necessitate cost-effective solutions.

The high rating for mobile accessibility (4.38) and low-internet functionality (4.28) reinforces the necessity of a mobile-first design with offline or low-bandwidth capabilities — consistent with the survey finding that 73.08% of respondents prefer accessing the system via their personal smartphones. The requirement for a simple, non-technical interface (4.43) is likewise critical given that the user base includes residents of varying educational backgrounds, including elementary graduates and older community members.

One respondent further elaborated: "Dapat igwa man an sistema nin automatic SMS alerts asin Facebook notifications tanganing dai mahuli an pag-abot kan impormasyon sa mga residente. Magayon man kun may Bikol asin Filipino language option para mas madaling masabutan kan gabos lalo na kan mga senior citizen. Mahalaga man na igwa nin backup battery o generator tanganing padagos an pag-andar kan sistema maski may brownout." (Translation: The system should also have automatic SMS alerts and Facebook notifications so that information reaches residents on time. It would be better if there is a Bikol and Filipino language option for easier comprehension especially for senior citizens. It is also important that there is a backup battery or generator so the system continues to function even during power outages.) This qualitative insight reinforces the importance of multilingual support, multi-channel notifications, and backup power provision in the system design.

3.6 Visualization and Interface Preferences

3.6.1 Preferred Visual Displays

Table 3.12: Preferred Visual Display Formats (Multi-Select)

Visual Display Format	Freq. (f)	Pct. (%)
Color-coded flood alert level map (Green/Yellow/Orange/Red zones)	69	88.46
Real-time water level gauge or meter display	55	70.51
Evacuation route map with highlighted safe paths	47	60.26
Line graph showing rising/falling water level trends	37	47.44
Weather forecast icons with rainfall prediction	35	44.87
Dashboard summary with key numbers (level, alert status, time to threshold)	34	43.59
Flood risk heat map showing most vulnerable areas	31	39.74
Text-based alerts with simple language (no charts needed)	28	35.90
Bar chart comparing current vs. historical flood levels	18	23.08

Table 3.12 presents the respondents' preferred visual display formats for the proposed system. The color-coded flood alert level map using the standard

Green/Yellow/Orange/Red alert system was the most preferred, selected by 69 respondents (88.46%). Real-time water level gauge displays ranked second at 70.51%, followed by evacuation route maps (60.26%). Line graphs for water level trends (47.44%), weather forecast icons (44.87%), and dashboard summaries (43.59%) were also highly preferred. These preferences indicate that the system's interface should prioritize spatial and color-coded visual indicators that allow users to quickly assess flood risk levels at a glance. The strong preference for evacuation route maps (60.26%) further suggests that the system should incorporate actionable guidance rather than merely providing raw data.

3.6.2 Helpfulness Ratings of Specific Visualization Types

Table 3.13: Helpfulness Ratings of Visualization Formats (5-Point Scale)

Visualization Format	**Wtd. Mean**	**Interpretation**
Color-coded flood level map (Green/Yellow/Orange/Red)	4.64	Extremely Helpful
Rainfall intensity display with forecast	4.45	Extremely Helpful
Graphs of predicted water level (3–12 hours ahead)	4.40	Extremely Helpful
Evacuation route overlay on barangay map	4.40	Extremely Helpful
Simple dashboard (current alert level in large text)	4.38	Extremely Helpful
Pop-up notifications with recommended actions	4.36	Extremely Helpful
Real-time water level gauge with threshold markers	4.21	Extremely Helpful
Animated map (expected flood spread/depth)	4.18	Very Helpful

Table 3.13 shows the helpfulness ratings for eight specific visualization formats. The color-coded flood level map received the highest weighted mean of 4.64, with 94% of respondents rating it as very or extremely helpful. Rainfall intensity displays with forecasts followed at 4.45 (95%), while predicted water level graphs (4.40) and evacuation route overlays (4.40) were equally rated, with 90% and 85% of respondents finding them very or extremely helpful respectively. All eight formats were rated above 4.00, indicating universal appreciation for diverse, data-rich visualization approaches in a flood monitoring system.

Table 3.14: Preferred Device for Accessing Flood Forecasting Information (Multi-Select)

Preferred Device	**Freq. (f)**	**Pct. (%)**
Smartphone (personal)	57	73.08
Any available device	35	44.87
Large public display screen at the barangay hall	18	23.08
Desktop or laptop computer	13	16.67
Tablet	13	16.67

3.6.3 Preferred Device for Accessing the System

Table 3.14 reveals that the personal smartphone is the most preferred device for accessing flood monitoring information ($n = 57$, 73.08%), followed by willingness to use any available device (44.87%). A significant portion of respondents (23.08%) also expressed the need for a large public display at the barangay hall, suggesting demand for a centralized information hub accessible even to residents without personal devices.

The high preference for smartphones reinforces the system design requirement for full mobile compatibility (Android and iOS), while the desire for a public display screen suggests a dual-access design — one for personal mobile use and another for community-level viewing in public spaces.

3.7 Summary of Findings, Conclusions and Recommendations

This presents the summary of findings conducted by the study as well as conclusions and recommendations.

3.7.1 Summary of Findings

The study aimed to design and develop a data-driven flood monitoring and forecasting system (FMFS) for Barangay Triangulo, Naga City, with the goal of addressing gaps in the barangay's existing manual flood monitoring practices. A total of 78 respondents participated in the survey, composed of 66 barangay residents (84.61%) and 12 barangay officials (15.39%). The majority belonged to the 18–35 age group (75.65%), suggesting a population generally receptive to technology-based flood monitoring solutions.

Key findings revealed that the barangay's flood monitoring remains predominantly manual and reactive. Manual visual inspection and physical patrols (73.08%) served as the most widely used monitoring method, followed by social media monitoring via Facebook (55.13%) and manual river water level observation (46.15%). Only 7.69% of respondents reported the use of automated sensors or IoT devices, and 5.13% noted the complete absence of any formal monitoring system. Monitoring activities were largely event-driven, with 41.03% of respondents indicating that monitoring is conducted only when typhoon or heavy rain warnings are issued.

Regarding the Flood Early Warning System (FEWS), 42.31% of respondents acknowledged that although one exists, it is not fully operational, while only 38.46% considered it fully functional. Flood warnings are primarily communicated through megaphones and public announcements (67.95%), Facebook (58.97%), and SMS (39.74%). Budget constraints were identified as the most significant challenge ($WM = 3.11$), alongside over-reliance on manual human effort ($WM = 3.04$), outdated equipment ($WM = 2.87$), communication breakdowns ($WM = 2.83$), and slow warning dissemination ($WM = 2.83$).

In terms of system requirements, respondents strongly supported the development of a comprehensive, data-driven FMFS. All twelve proposed functional features were rated in the Extremely Important range (WM: 4.28–4.52), with automated flood alerts and real-time water level monitoring receiving the highest ratings at 4.52 each. All ten non-functional requirements were likewise rated Extremely Important (WM: 4.28–4.53), led by backup power capability and automated DRRM report generation (4.53 each).

For visualization preferences, the color-coded flood alert level map (Green/Yellow/Orange/Red) was the most preferred display format (88.46%) and received the highest helpfulness rating at 4.64. Real-time water level gauges (70.51%) and evacuation route maps (60.26%) also ranked highly. Personal smartphones were the most preferred device for accessing the system (73.08%), reinforcing the need for a mobile-first design with offline and low-bandwidth capabilities.

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